INTEGRATIVE PROJECT · DATA MINING · MACHINE LEARNING · CLOUD COMPUTING

TRM Predict

A cloud-native predictive analytics system for exchange-rate risk decisions in Colombia

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Data Engineering and Artificial Intelligence, 5th semester · Universidad Santo Tomás, Tunja · 2026-2

Code, data and Terraform

2,159 trading days, 43 leakage-free features

19 models tuned with time-series CV

0.586 test ROC-AUC for large-move risk

+383 COP/day per USD 1,000 for a remittance household

100% availability while a server is killed

+84% throughput after auto-scaling 2 → 4

1 The problem

Colombian households received about **USD 11.8 billion** in remittances in 2024, and thousands of small importers pay in dollars. Both convert at the official exchange rate (**TRM**) without the analytics of large firms.

Question: can public market data say, one business day ahead, whether the TRM will rise and whether a large move is coming, and can that be served reliably and cheaply in the cloud?

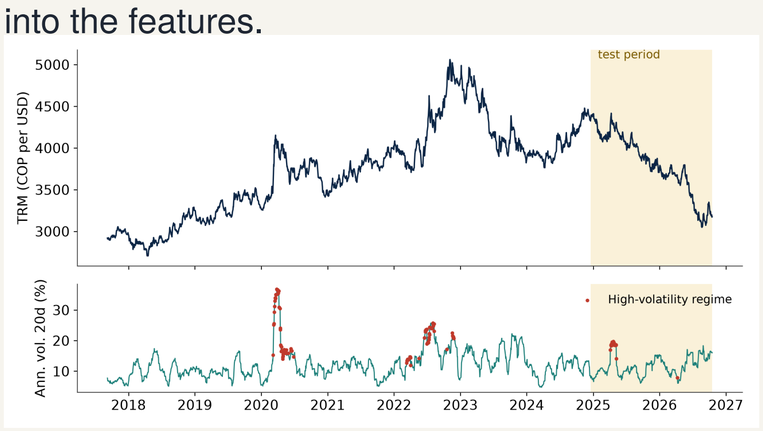
Aligned with SDG 8 (target 8.10, access to financial services) and SDG 10 (target 10.c, remittance costs).

2 Data and pipeline

A CRISP-DM pipeline integrates four sources, refreshed every weekday by a serverless job:

- Official TRM (datos.gov.co API, since 1991)
- 8 markets: Brent, DXY, US 10Y, VIX, S&P 500, MXN, BRL, spot USD/COP (Yahoo)
- BTC and ETH daily candles (Binance)
- FX news sentiment, **web-scraped** from *La República*

A strict as-of join uses only market closes *before* each TRM date, so no future information leaks into the features.



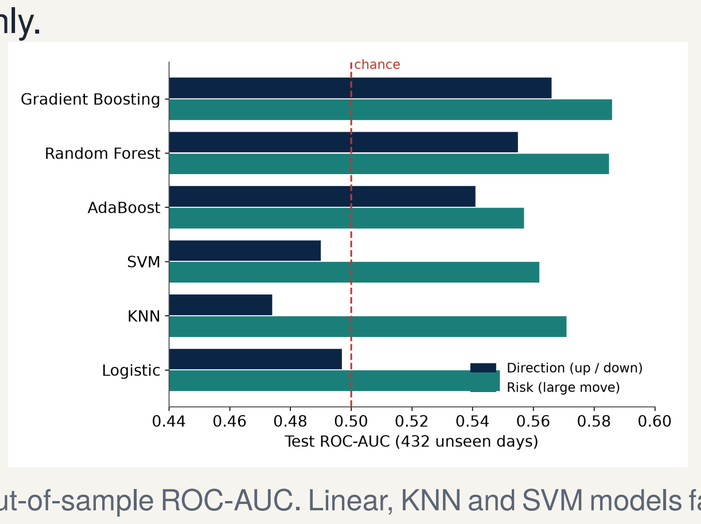
TRM and 20-day volatility, Sep. 2017 to Oct. 2026. Red: days in the high-volatility regime found by clustering. Shaded: the 432 test days, never used for training.

3 Data mining

- **PCA + K-Means / Ward:** 3 market regimes. In risk-off days the TRM rises next day 52.9% of the time; in risk-on days, 44.3%.
- **DBSCAN** flags **57 anomalous days** (45 in 2020), with 1.7× the normal next-day volatility.
- **Apriori = FP-Growth:** 91 rules, e.g. *BRL up & MXN up & VIX up* ⇒ *TRM up* (confidence 0.60, lift 1.24).

4 Machine learning

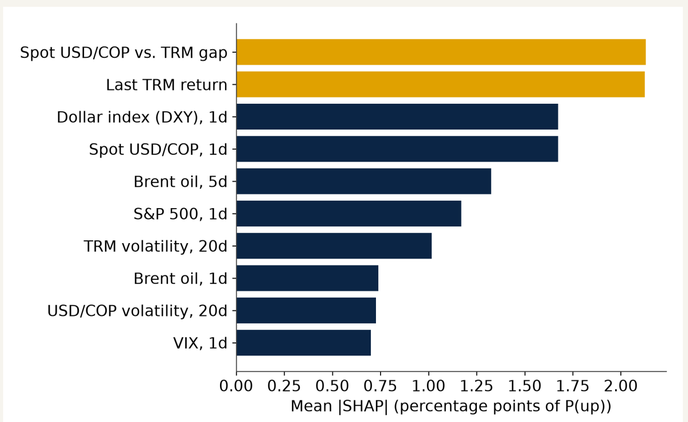
Three tasks for the next business day: **direction** (up/down), **risk** (move larger than the 75th percentile, 0.86%) and **return** (regression). 19 models were tuned with Grid/Random Search inside a 5-fold `TimeSeriesSplit` and chosen by CV score only.



Out-of-sample ROC-AUC. Linear, KNN and SVM models fall to chance; only tree ensembles keep a signal.

- Regression: Gradient Boosting RMSE **2.4% below the random walk**; Random Forest gets the sign right 54.6% of the time.
- Selected: **Random Forest** for all three tasks (best CV in 2 of 3, one family to maintain).

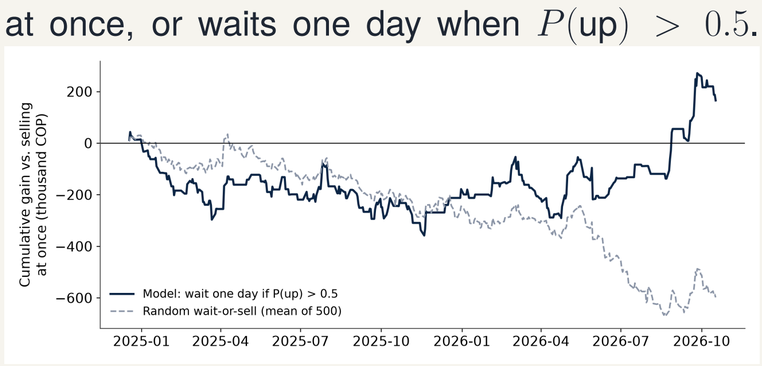
5 Why the model decides



TreeSHAP importance. The engineered spot–TRM gap ranks first: the TRM averages the day's trades, so a spot quote above it anticipates a rise.

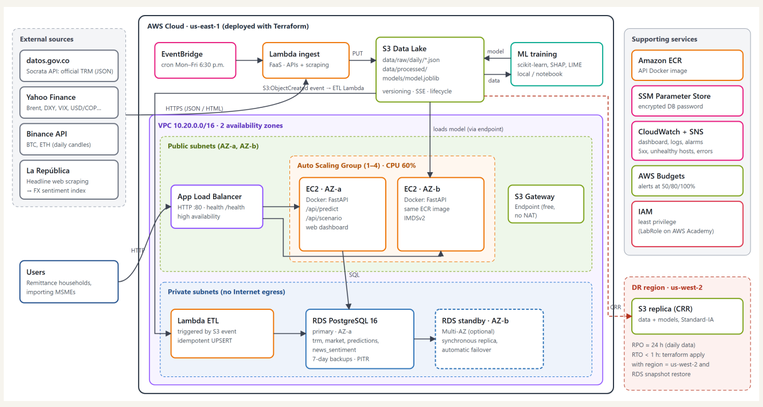
6 Does it help a user?

A household receiving USD 1,000 a day sells at once, or waits one day when $P(\text{up}) > 0.5$.



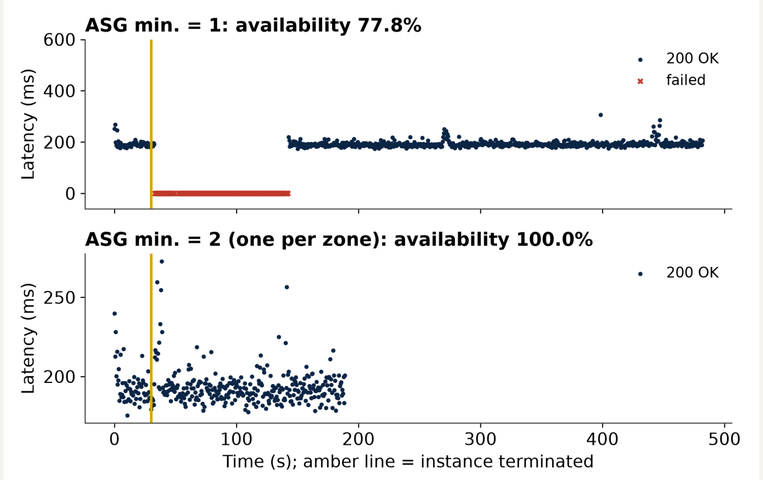
Test period. Model: +383 COP/day (+624 with threshold 0.55). Random choice: −1,383 COP/day. The risk alert has 30.1% precision vs. a 22.5% base rate (lift 1.34).

7 Architecture on AWS

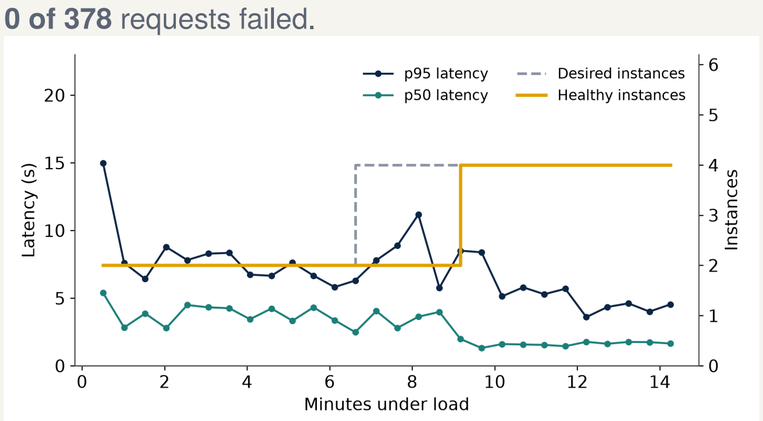


64 resources in Terraform. Lambda ingestion and ETL, S3 data lake replicated to us-west-2, RDS PostgreSQL, and a Docker FastAPI behind an ALB with an Auto Scaling group across two availability zones.

8 Cloud experiments



One instance terminated at $t = 30$ s. With an ASG minimum of 1 the site was down for 109 s; with one instance per zone, **0 of 378** requests failed.



16 concurrent clients for 14 min. Target tracking at 60% CPU scales 2 → 4: 4.1 → 7.6 req/s, median latency 4.0 → 1.6 s, **0 errors in 4,543** requests.

9 Cost and conclusions

About **USD 64/month** on demand (ALB, EC2, public IPv4 and RDS dominate); no NAT gateway saves USD 33. Created and destroyed with Terraform, the whole semester costs about **USD 21**.

- Daily FX is hard to predict, but a small, honest edge can still beat random decisions.
- High availability needs the right *scaling policy*, not only two zones.

References: Meese & Rogoff (1983) · Breiman (2001) · Friedman (2001) · Lundberg & Lee (2017) · Ester et al. (1996) · Agrawal & Srikant (1994)